Cloud Analytics Project Setup & ETL Cheat Sheet

PROJECT OVERVIEW

Goal:

Build an end-to-end cloud analytics platform using:

- CSV/Text Files
- Google Cloud Storage (GCS)
- Python ETL
- BigQuery
- Power BI

Final Architecture:

```
CSV Files
    ↓
Google Cloud Storage
    ↓
Python ETL Pipeline
    ↓
BigQuery
    ↓
Power BI
```

STEP 1 — CREATE PROJECT FOLDER

Example:

```
C:\olist-ecommerce-analytics
```

STEP 2 — CREATE PROJECT STRUCTURE

```
olist-ecommerce-analytics/
│
├── data/
```

```
│   ├── raw/
│   ├── processed/
│   └── archive/
│
├── notebooks/
├── scripts/
├── sql/
├── screenshots/
├── powerbi/
├── credentials/
│
├── README.md
├── requirements.txt
├── .gitignore
```

STEP 3 — OPEN TERMINAL INSIDE PROJECT FOLDER

Example:

```
cd C:\olist-ecommerce-analytics
```

STEP 4 — CREATE VIRTUAL ENVIRONMENT

```
python -m venv venv
```

STEP 5 — ACTIVATE VIRTUAL ENVIRONMENT

CMD

```
venv\Scripts\activate
```

PowerShell

```
.\venv\Scripts\Activate.ps1
```

When activated you should see:

```
(venv)
```

STEP 6 — INSTALL REQUIRED PACKAGES

```
pip install pandas google-cloud-storage google-cloud-bigquery pandas-gbq
pyarrow jupyter
```

STEP 7 — SAVE REQUIREMENTS

```
pip freeze > requirements.txt
```

STEP 8 — CREATE .gitignore FILE

Paste this:

```
venv/
credentials/
__pycache__/
*.pyc
.env
.ipynb_checkpoints/
```

STEP 9 — CREATE GCP BUCKET

Recommended settings:

Setting	Value
Location Type	Region
Region	us-central1
Storage Class	Standard
Access Control	Uniform
Public Access Prevention	Enabled

Example bucket:

```
cdmx_mobility_insight2024
```

STEP 10 — CREATE BIGQUERY DATASET

Example:

```
rides_2024
```

STEP 11 — CREATE SERVICE ACCOUNT

Roles:

- Storage Admin
- BigQuery Admin

Download JSON key.

Store here:

```
credentials/olist-gcp.json
```

STEP 12 — SET GOOGLE CREDENTIALS

Option 1 — Environment Variable (CMD)

```
set GOOGLE_APPLICATION_CREDENTIALS=C:\olist-ecommerce-
analytics\credentials\olist-gcp.json
```

Option 2 — Inside Python Script

```
import os

os.environ["GOOGLE_APPLICATION_CREDENTIALS"] = (
    "credentials/olist-gcp.json"
)
```

STEP 13 — START JUPYTER NOTEBOOK

```
jupyter notebook
```

STEP 14 — TEST GCS CONNECTION

```
from google.cloud import storage

client = storage.Client()

buckets = list(client.list_buckets())

for bucket in buckets:
    print(bucket.name)
```

STEP 15 — UPLOAD FILE TO GCS

```
from google.cloud import storage

PROJECT_ID = "cdmxmobility"
BUCKET_NAME = "cdmx_mobility_insight2024"

FILE_PATH = "data/processed/final_orders.csv"
DEST_BLOB_NAME = "final_orders.csv"

storage_client = storage.Client(project=PROJECT_ID)

bucket = storage_client.bucket(BUCKET_NAME)
blob = bucket.blob(DEST_BLOB_NAME)

blob.upload_from_filename(FILE_PATH)

print("File uploaded successfully")
```

STEP 16 — LOAD CSV INTO BIGQUERY

```
from google.cloud import bigquery

PROJECT_ID = "cdmxmobility"
DATASET_ID = "rides_2024"
TABLE_ID = "final_orders"
BUCKET_NAME = "cdmx_mobility_insight2024"

uri = f"gs://{BUCKET_NAME}/final_orders.csv"

client = bigquery.Client(project=PROJECT_ID)

table_ref = f"{PROJECT_ID}.{DATASET_ID}.{TABLE_ID}"

job_config = bigquery.LoadJobConfig(
    source_format=bigquery.SourceFormat.CSV,
    skip_leading_rows=1,
    autodetect=True,
    write_disposition="WRITE_TRUNCATE"
)

load_job = client.load_table_from_uri(
```

```
    uri,
    table_ref,
    job_config=job_config
)

load_job.result()

print("Data loaded successfully")
```

STEP 17 — CREATE RFM TABLE

```
import pandas as pd

final_orders = pd.read_csv(
    "data/processed/final_orders.csv"
)

final_orders['order_purchase_timestamp'] = pd.to_datetime(
    final_orders['order_purchase_timestamp']
)

snapshot_date = (
    final_orders['order_purchase_timestamp'].max()
    + pd.Timedelta(days=1)
)

rfm = final_orders.groupby('customer_unique_id').agg({
    'order_purchase_timestamp': lambda x: (snapshot_date - x.max()).days,
    'order_id': 'nunique',
    'payment_value': 'sum'
}).reset_index()

rfm.columns = [
    'customer_unique_id',
    'Recency',
    'Frequency',
    'Monetary'
]

print(rfm.head())
```

STEP 18 — EXPORT RFM TABLE

```
rfm.to_csv(
    'data/processed/final_orders_rfm.csv',
    index=False
)
```

STEP 19 — LOAD RFM TABLE INTO BIGQUERY

```
TABLE_ID = "final_orders_rfm"

uri = f"gs://{BUCKET_NAME}/final_orders_rfm.csv"
```

Repeat the same BigQuery loading process.

STEP 20 — CONNECT POWER BI TO BIGQUERY

In Power BI:

```
Get Data
    ↓
Google BigQuery
```

Select:

```
cdmxmobility
```

Then:

```
rides_2024
```

Import:

- final_orders
- final_orders_rfm

STEP 21 — BUILD DASHBOARDS

Recommended pages:

1. Executive Overview

Focus:

- revenue
- customers
- orders
- sales trends

---

2. Logistics Dashboard

Focus:

- delivery delays
- shipping performance
- operational efficiency

---

3. RFM Dashboard

Focus:

- customer segmentation
- retention
- loyalty
- customer value

STEP 22 — SAVE POWER BI FILE

Example:

```
powerbi/olist_ecommerce_analytics.pbix
```

STEP 23 — GIT COMMANDS

Initialize Git

```
git init
```

Add Files

```
git add .
```

Commit

```
git commit -m "Initial commit"
```

Connect GitHub Repo

```
git remote add origin YOUR_REPO_URL
```

Push to GitHub

```
git branch -M main

git push -u origin main
```

STEP 24 — AUTOMATION (NEXT LEVEL)

Future automation workflow:

```
CSV Uploaded to GCS
        ↓
```

```
Cloud Function Triggered
        ↓
Python ETL Runs
        ↓
BigQuery Updated
        ↓
Power BI Refreshes
```

Tools:

- Cloud Functions
- Cloud Scheduler
- Cloud Run

FINAL PORTFOLIO SKILLS

This project demonstrates:

- Python ETL
- Cloud Storage
- BigQuery
- SQL Analytics
- Power BI
- Customer Analytics
- Logistics Analytics
- Cloud Architecture
- Automation Workflow
- GitHub Project Structuring

RECOMMENDED PROJECT TITLE

```
Cloud-Based E-Commerce Analytics Platform
```

or

```
End-to-End Cloud Analytics Pipeline with BigQuery & Power BI
```

FINAL CAREER VALUE

This project is suitable for:

- Data Analyst Portfolio
- BI Analyst Portfolio
- Analytics Engineer Portfolio
- Junior Data Engineer Portfolio

because it combines:

- analytics
- engineering
- cloud
- automation
- business intelligence
- customer analytics